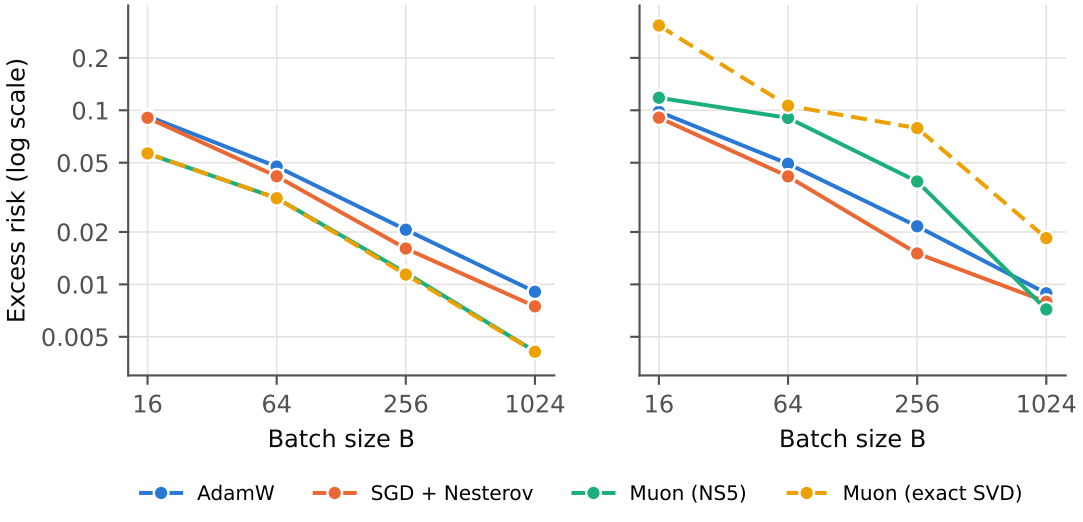
(a) Equal steps: 6000 per run
(b) Equal time: SGD's 6000-step time
Excess risk (log scale)
0.2
0.1
0.05
0.02
0.01
0.005
16
64
256
1024
Batch size B
AdamW
SGD + Nesterov
Muon (NS5)
Muon (exact SVD)